P16 · LSTM vs Transformer Coherence Test

Evaluating Long-Range Story Continuity on TinyStories (Children's Stories)

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Course: Generative AI (Mini-Project)	**Problem Statement:** P16 (Track B)
Corpus: TinyStories (Preschool Vocabulary)	**Live App:** https://p16-coherence-1047195478355.us-central1.run.app

1. Problem Statement & Objective

The Core Question: Why did Transformers replace Recurrent Neural Networks (LSTMs) for text generation?
To answer this without the distracting complexity of archaic English, we compare a **character-level LSTM (Keras/TF2)** against a **distilGPT-2 Transformer (PyTorch)** on **TinyStories**—a dataset of simple children's stories about dogs, balls, friends, and snacks. Because children's stories have clear, simple plots (e.g. *'Tim found a puppy... the puppy ran into the woods'*), any failure to track character names or actions becomes immediately obvious to human readers and automated judges alike.

2. Full Syllabus Coverage (Units U1–U5)

Unit	Requirement	Implementation Details
U1 Foundations	Perplexity & Shared Sampling	Fixed sampling (T=0.7, top-k=40, top-p=0.9). Character-normalised perplexity computed on identical held-out stories.
U2 GANs & LLMs	LSTM vs Transformer internals	Trained 2-layer LSTM from scratch; fine-tuned distilGPT-2. Extracted layer-wise attention distance showing how far back Transformers attend.
U3 Models	GPT + BERT Judge	BERT Next-Sentence-Prediction (NSP) scores coherence between sentence pairs separated by distance d=1 to d=5.
U4 Tools & Gen	Keras + PyTorch + TTS	Hybrid Keras & PyTorch architecture. Reads the winning, most coherent story aloud using Google Text-to-Speech (gTTS).
U5 Responsible AI	Memorisation & Bias Guardrails	Scans for verbatim 8-gram copying from training data. Pronoun representation probe across story roles (friend, helper, doctor).

3. Side-by-Side Comparison on Simple Story Prompts

Prompt: *'Once upon a time, there was a little boy named Tim. Tim had a pet dog named'*

char-LSTM (Keras)	distilGPT-2 (Transformer)

'...Spot. Spot was a good dog and he like to play with Tim. But one day the dag saw a big trree and ron to the gardn. Tim cried for his pet and then he saw a cat on the flor...' **Observation:** The LSTM invents misspellings ('dag', 'trree', 'ron') and forgets the puppy Spot to suddenly talk about a cat. Long-range memory vanishes.	*'...Max. Max loved to run in the green park with Tim. They played with a little yellow ball. When the ball rolled under a bush, Max barked happily and brought it back to Tim. Tim gave Max a big hug.'* **Observation:** Flawless syntax, consistent character tracking (Tim and Max remain friends throughout), and high story coherence.

4. Quantitative Experimental Results

Metric	char-LSTM (Keras)	distilGPT-2 (Base)	distilGPT-2 (TinyStories FT)
Validation Char Perplexity ↓	3.42	3.18	**2.45**
Bits Per Character (BPC) ↓	1.77	1.67	**1.29**
Local Coherence (BERT NSP, d=1) ↑	0.84	0.89	**0.97**
Long-Range Coherence (BERT NSP, d≥3) ↑	0.38 (Severe Drop)	0.71	**0.91 (Maintained)**
Verbatim 8-gram Copied % ↓	0.4%	0.0%	1.2%

5. Documented Failure Cases & Ethical Review

Documented Failure Case: When prompted with repetitive inputs (e.g. *'The cat and the cat and the cat'*), the LSTM enters an infinite repeating loop, producing *'and the cat and the cat...'* until token cutoff. The Transformer breaks out of the loop after two repetitions due to its global multi-head self-attention.

Responsible AI & Ethics:
• **Memorisation:** DistilGPT-2 exhibited a 1.2% rate of 8-gram replication on highly repetitive nursery formulas (e.g. *'Once upon a time there was a'*). It did not memorise unique story plots.
• **Safety Guardrails:** The app includes regex-based safety screening for PII redaction and blocks inappropriate/harmful content.
• **Licence:** TinyStories is released under open terms; DistilGPT-2 is Apache 2.0; code is licensed under MIT.

6. Conclusion

This experiment demonstrates why Transformers replaced LSTMs: while LSTMs achieve passable local character transitions, their fixed-size hidden state bottlenecks long-range memory, causing story plots and character identities to decay rapidly after 2–3 sentences. Transformers resolve this through full cross-token attention, maintaining high coherence across entire paragraphs.